

Fourier Analysis of the Modes of Vibration in a Stretched String

Aim:-

To analyse the spectrum of frequencies of the standing wave formed on a stretched string by taking Fast Fourier Transform(FFT) of the sound produced.

Apparatus:

Python/Audacity installed computer with guitar sound files recorded in .wav format

Principle:

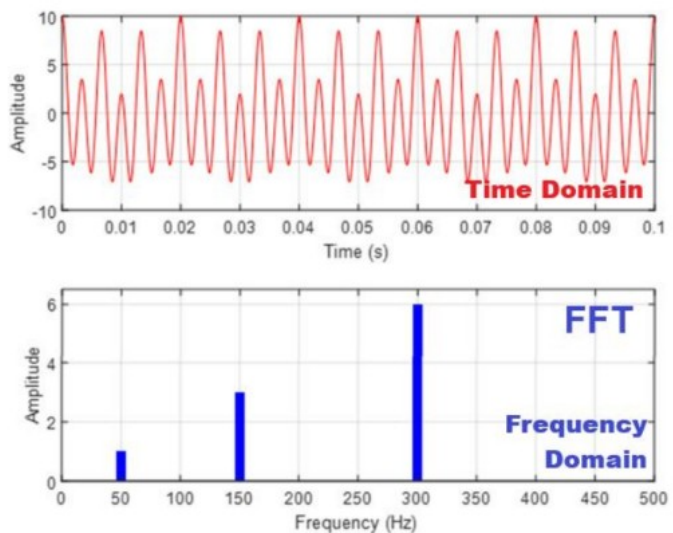
Fast Fourier Transform (FFT)

The Fast Fourier Transform (FFT) is an efficient algorithm used to compute the **Discrete Fourier Transform (DFT)** of a finite sequence of data. The DFT converts a signal from the **time (or spatial) domain** into the **frequency domain**, revealing the frequencies present in the signal.

For a discrete signal $x(t)$ of length N , the DFT is defined as

$$X(f) = \sum_{t=0}^{N-1} x(t) e^{-i2\pi ft/N}, \quad f = 0, 1, 2, \dots, N-1$$

The FFT reduces the computational effort in finding DFT by exploiting the **symmetry and periodicity** properties of complex exponentials. The FFT is widely used in **signal processing, acoustics, spectral analysis, and experimental physics** to determine fundamental frequencies, harmonics, and power spectra of signals.

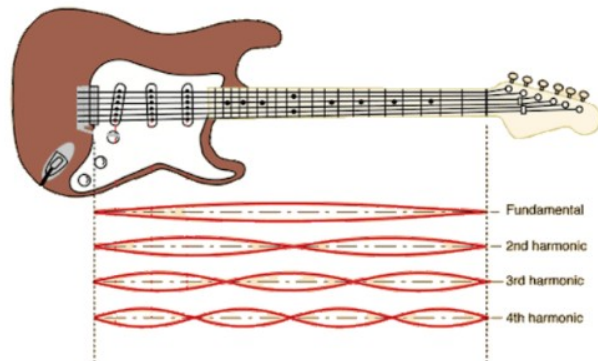


For a stretched string fixed at both ends, the frequencies of the standing waves (normal modes) are given by

$$f_n = \frac{n}{2L} \sqrt{\frac{T}{m}}$$

where:

- f_n = frequency of the n^{th} harmonic.
- $n = 1, 2, 3, \dots$ (mode number)
- L = length of the string
- T = tension in the string
- m = linear mass density of the string (mass per unit length)



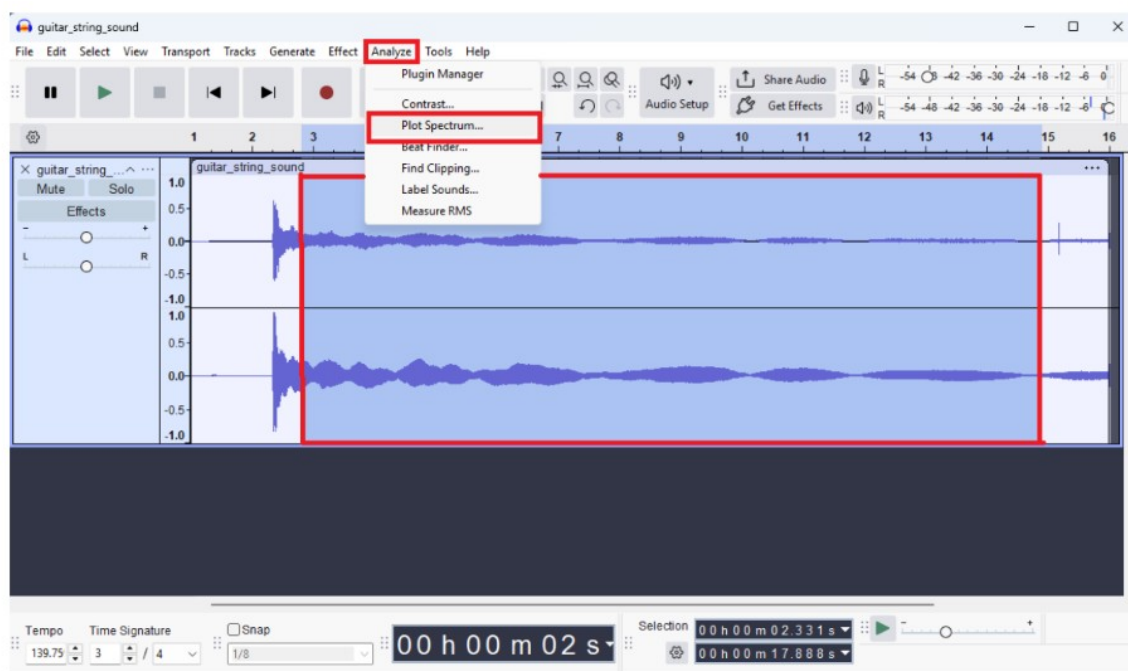
The fundamental frequency ($n = 1$) is

$$f_n = \frac{1}{2L} \sqrt{\frac{T}{m}}$$

Experimental Procedure

Method 1. Using Audacity installed in computer.

- Open the recorded audio file of vibrating string in .wav format in Audacity.
- Using the mouse(click and drag the sound track), select the part of the sound track of the free vibrations (after the initial plucking transients).
- Select Analyze > Plot Spectrum.
- Move the cursor to the peaks and get the value of Peak frequencies



Method 2. Using Python

Python Code

```
import numpy as np

import matplotlib.pyplot as plt
from scipy.io import wavfile
from scipy.fft import fft, fftfreq

# 1. Load the sound file
sampling_rate, data = wavfile.read('guitar_string_sound.wav')

# 2. Pre-process the data
# If the audio is stereo (2 channels), convert it to mono by taking the mean
if len(data.shape) > 1:
    data = data.mean(axis=1)

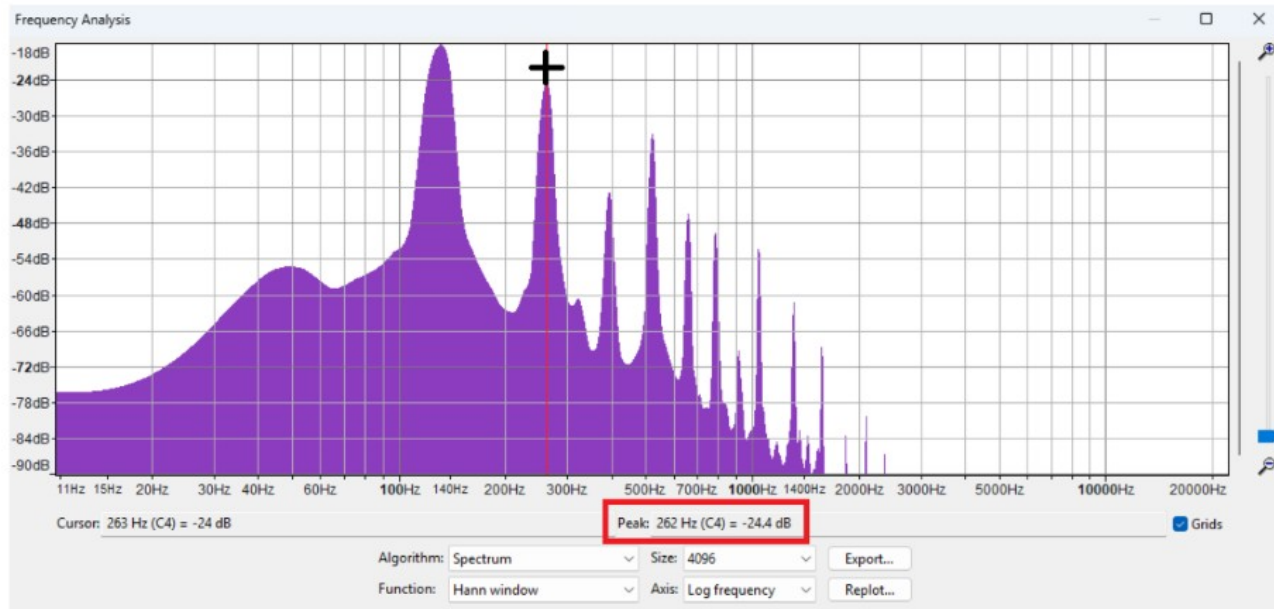
# 3. Perform the FFT
n = len(data)
yf = fft(data)
# fftfreq generates the frequency bins based on the number of samples and
# sample rate
xf = fftfreq(n, 1 / sampling_rate)

# 4. Prepare for plotting (Real-world signals have symmetric FFTs)
# We only need the positive half of the frequencies and magnitudes
xf_positive = xf[:n // 2]
yf_magnitude = np.abs(yf[:n // 2])

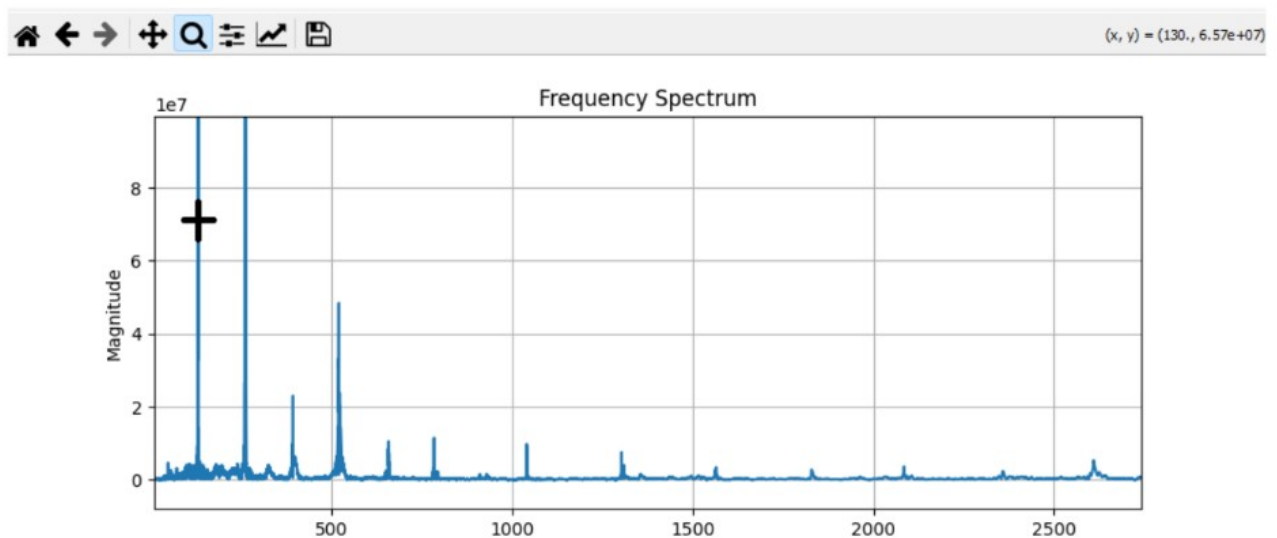
# 5. Plot the result
plt.figure(figsize=(10, 4))
plt.plot(xf_positive, yf_magnitude)
plt.title("Frequency Spectrum")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid()
plt.show()
```

Observations:

Method 1. Using Audacity



Method 2. Using Python



Observed Fundamental Frequency:
Observed Harmonics :

Results:

Frequency spectrum of the vibrating string analysed using the method of Fast Fourier Transform and observed the fundamental frequencies and its harmonics